# Predictive Analytics Capstone

## Part 6 – Store Segmentation, Classification, and Timeseries Forecasting


ABSTRACT

Throughout this report, I will demonstrate every taken step to came up with a segmentation solution to help my company remedy the product surplus and shortages, determine new stores formats, and forecasting Produce Sales for 2016.



Maged Baheig

**Problem Solving with Advanced Analytics-Udacity/Bertelsmann2021**

## *Predictive Analytics Capstone*

*Complete each section. When you are ready, save your file as a PDF document and submit it here:*
https://coco.udacity.com/nanodegrees/nd008/locale/en-us/versions/1.0.0/parts/7271/project

# Task 1: Determine Store Formats for Existing Stores

*Answer the following questions to help you plan out your analysis:*

1. What is the optimal number of store formats? How did you arrive at that number?

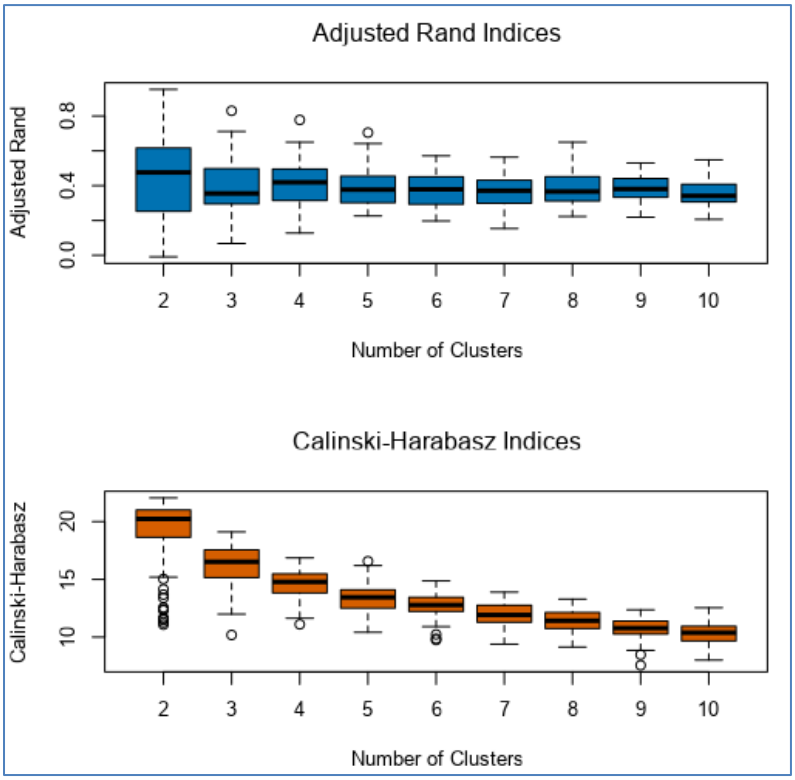
- The Optimal number of store format is: **Three Formats (3 Segments)**.

- I arrived that number via: **K-Centroids Diagnostics Tool** at Alteryx.

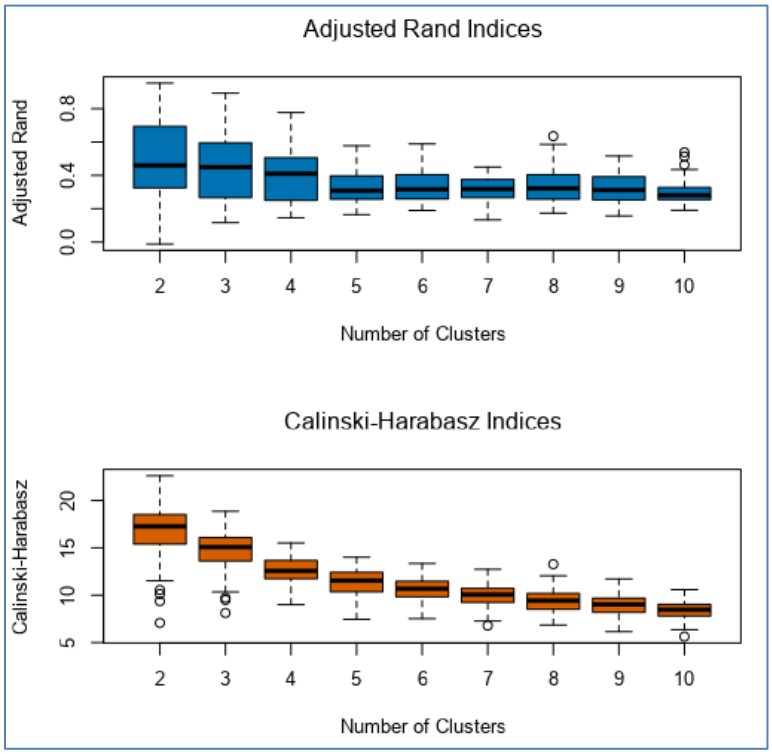
  - After running 3 clustering models (**K-means**, **K-median**, **Neural Gas**), I decided to use the **K-means** clustering method.

  - Using the **median** and **spread** of the **Rand** and **CH** (Calinski-Harabasz) Indices. It's clear that **three clusters** the most **optimal** method because the box-whisker plots in both the **Rand** & **CH** indices show how **tight** the indices for each data point are within each other.

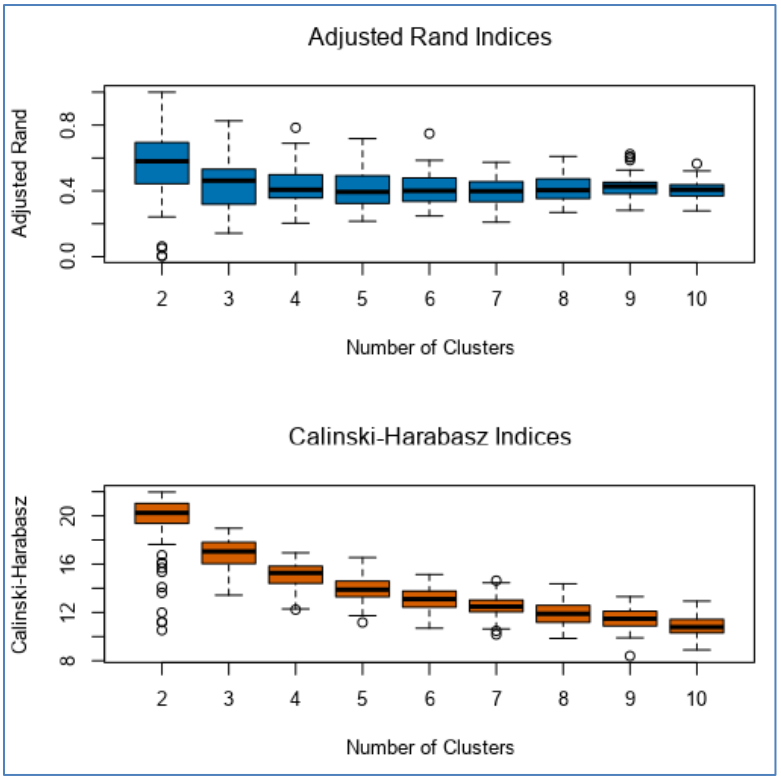
**K-Means Diagnostics**

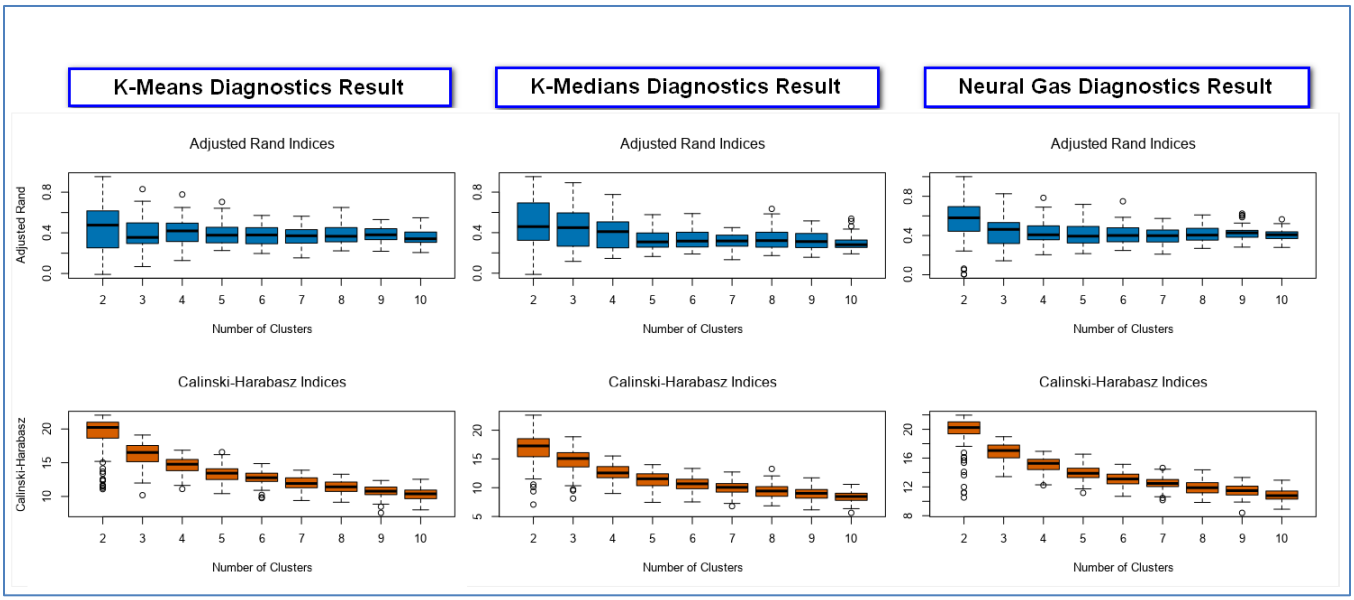


## K-Medians Diagnostics




## Neural Gas Diagnostics






- <u>From the **above** charts, and **below** statistical properties table</u>, we can see that:
  - The **Neural Gas** is better than **K-Means** and **K-Median** at Cluster **3,** especially for **<u>CH</u>** indices, regarding **Center measures** (*Higher Mean and Median)* and **Compactness measures** *(Lower Range & IRQ*).
  - **<u>However</u>**, I decided to use **K-Means** just because **UDACITY Knowledge Help** and **UDACITY Classroom** informed me to do that. **Please check below links**:
    - https://knowledge.udacity.com/questions/751444
    - https://knowledge.udacity.com/questions/245335
    - https://knowledge.udacity.com/questions/276672
  - **<u>But</u>**, if you look deeply at above **chart** and below **statistics**, you will be **highly confident** that; **Neural Gas is better than K-Means** and **K-Medians**.

**K-Means Cluster Assessment Report**

| Adjusted Rand Indices | 2 | 3 | 4 |
|---|---|---|---|
| Minimum | -0.008598 | 0.06782 | 0.127587 |
| 1st Quartile | 0.26367 | 0.295924 | 0.315603 |
| Median | 0.475963 | 0.355163 | 0.419058 |
| Mean | 0.448709 | 0.395469 | 0.399846 |
| 3rd Quartile | 0.607516 | 0.494664 | 0.495652 |
| Maximum | 0.95293 | 0.830305 | 0.777784 |
| **Range** | 0.961528 | 0.762485 | 0.650197 |
| **IRQ** | 0.343846 | 0.19874 | 0.180049 |

| Calinski-Harabasz Indices | 2 | 3 | 4 |
|---|---|---|---|
| Minimum | 11.08162 | 10.18405 | 11.10534 |
| 1st Quartile | 18.6546 | 15.20253 | 13.82227 |
| Median | 20.23875 | 16.51546 | 14.77552 |
| Mean | 19.14751 | 16.24817 | 14.6316 |
| 3rd Quartile | 21.00801 | 17.55597 | 15.47732 |
| Maximum | 22.05502 | 19.10661 | 16.86351 |
| **Range** | 10.9734 | 8.92256 | 5.75817 |
| **IRQ** | 2.35341 | 2.35344 | 1.65505 |

**K-Medians Cluster Assessment Report**

| Adjusted Rand Indices | 2 | 3 | 4 |
|---|---|---|---|
| Minimum | -0.011905 | 0.116351 | 0.145613 |
| 1st Quartile | 0.331248 | 0.269989 | 0.251812 |
| Median | 0.459513 | 0.449192 | 0.410309 |
| Mean | 0.497246 | 0.459494 | 0.395722 |
| 3rd Quartile | 0.694071 | 0.592027 | 0.505669 |
| Maximum | 0.952935 | 0.893389 | 0.777026 |
| **Range** | 0.96484 | 0.777038 | 0.631413 |
| **IRQ** | 0.362823 | 0.322038 | 0.253857 |

| Calinski-Harabasz Indices | 2 | 3 | 4 |
|---|---|---|---|
| Minimum | 7.072169 | 8.123626 | 8.996668 |
| 1st Quartile | 15.412619 | 13.622304 | 11.766219 |
| Median | 17.267148 | 15.07803 | 12.566029 |
| Mean | 16.775191 | 14.70207 | 12.526293 |
| 3rd Quartile | 18.507733 | 16.046106 | 13.661721 |
| Maximum | 22.607996 | 18.857375 | 15.514958 |
| **Range** | 15.535827 | 10.733749 | 6.51829 |
| **IRQ** | 3.095114 | 2.423802 | 1.895502 |

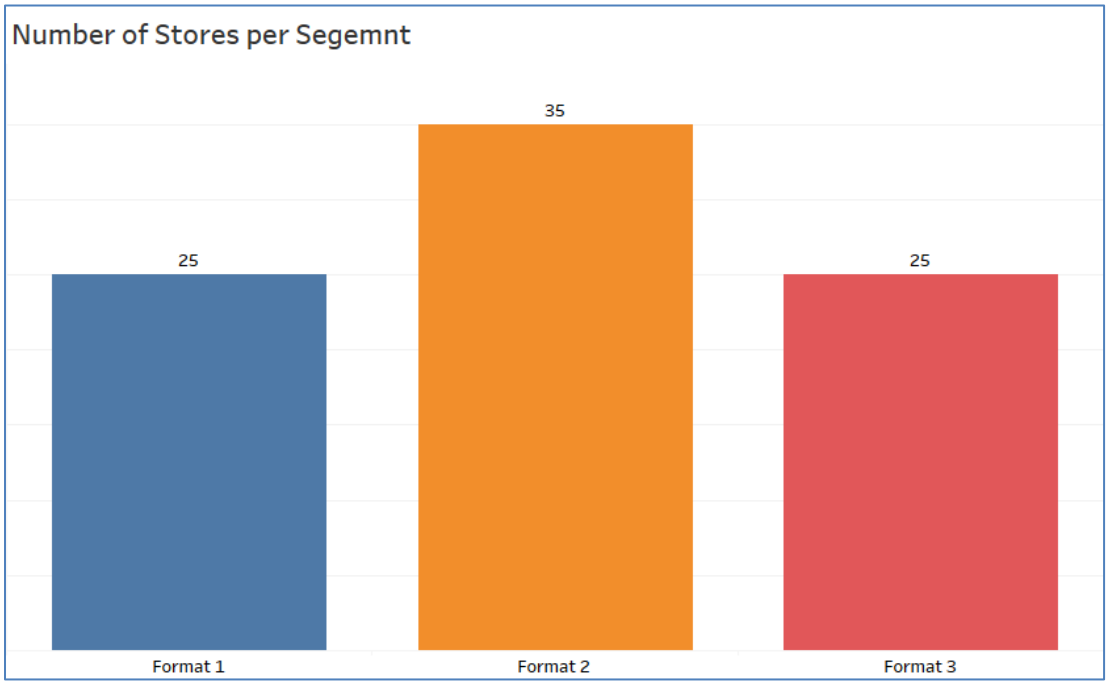
**Neural Gas Cluster Assessment Report**

| Adjusted Rand Indices | 2 | 3 | 4 |
|---|---|---|---|
| Minimum | 0.003827 | 0.142572 | 0.20361 |
| 1st Quartile | 0.443062 | 0.319949 | 0.357012 |
| Median | 0.579907 | 0.462438 | 0.407124 |
| Mean | 0.55567 | 0.449522 | 0.429847 |
| 3rd Quartile | 0.694147 | 0.532477 | 0.498159 |
| Maximum | 1 | 0.825766 | 0.784135 |
| **Range** | 0.996173 | 0.683194 | 0.580525 |
| **IRQ** | 0.251085 | 0.212528 | 0.141147 |

| Calinski-Harabasz Indices | 2 | 3 | 4 |
|---|---|---|---|
| Minimum | 10.52522 | 13.42627 | 12.22926 |
| 1st Quartile | 19.4428 | 16.04861 | 14.41511 |
| Median | 20.24233 | 17.04579 | 15.26792 |
| Mean | 19.64133 | 16.82188 | 15.09739 |
| 3rd Quartile | 21.01586 | 17.80189 | 15.82764 |
| Maximum | 21.96157 | 18.97202 | 16.94195 |
| **Range** | 11.43635 | 5.54575 | 4.71269 |
| **IRQ** | 1.57306 | 1.75328 | 1.41253 |

2. How many stores fall into each store format?

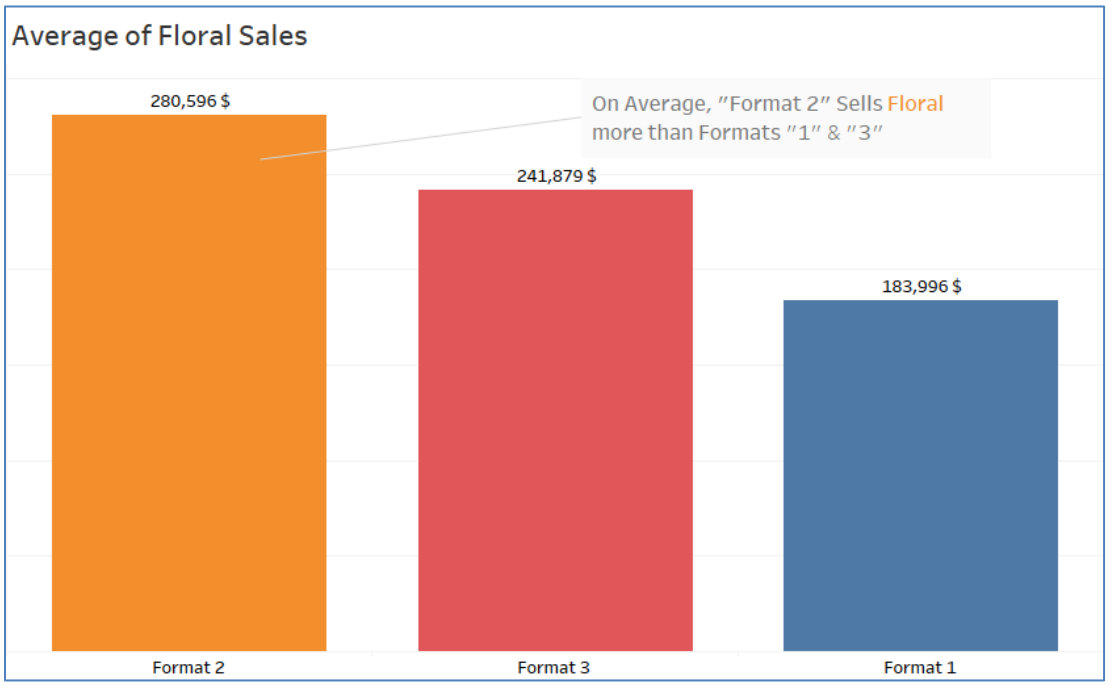
- Each store format has about: From **25** to **35 store per a segment**.
  - **Format 1**: **25** Stores.
  - **Format 2**: **35** Stores.
  - **Format 3**: **25** Stores.



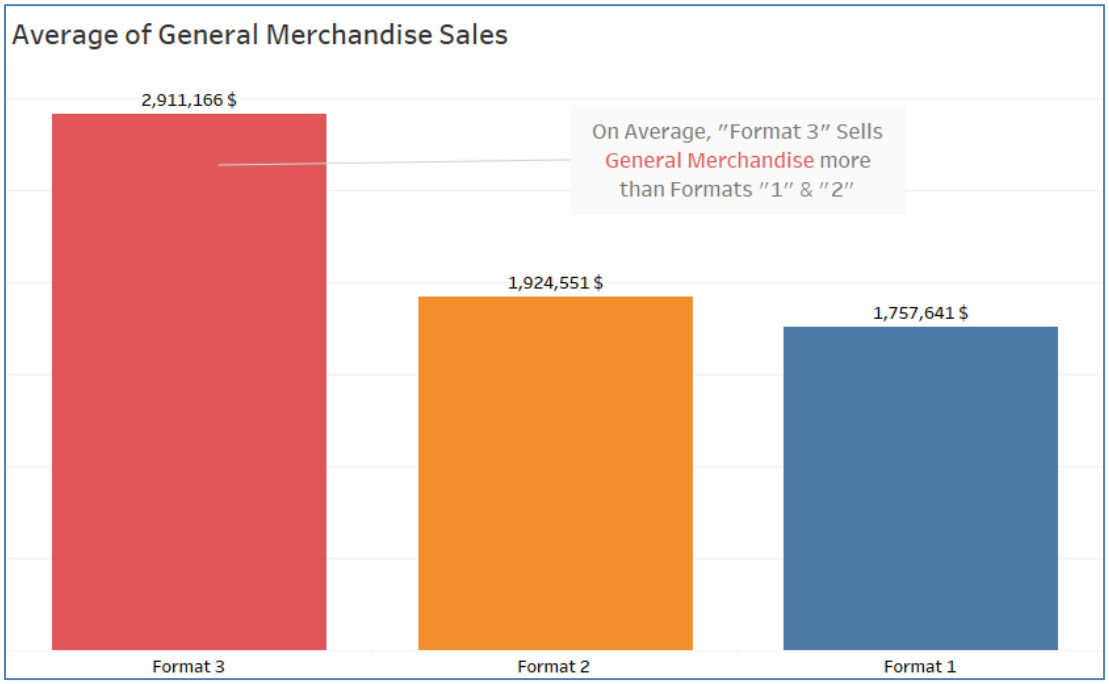
3. Based on the results of the clustering model, what is one way that the clusters differ from one another?

- <u>We can differentiate the clusters from one another by sales category, as below</u> :

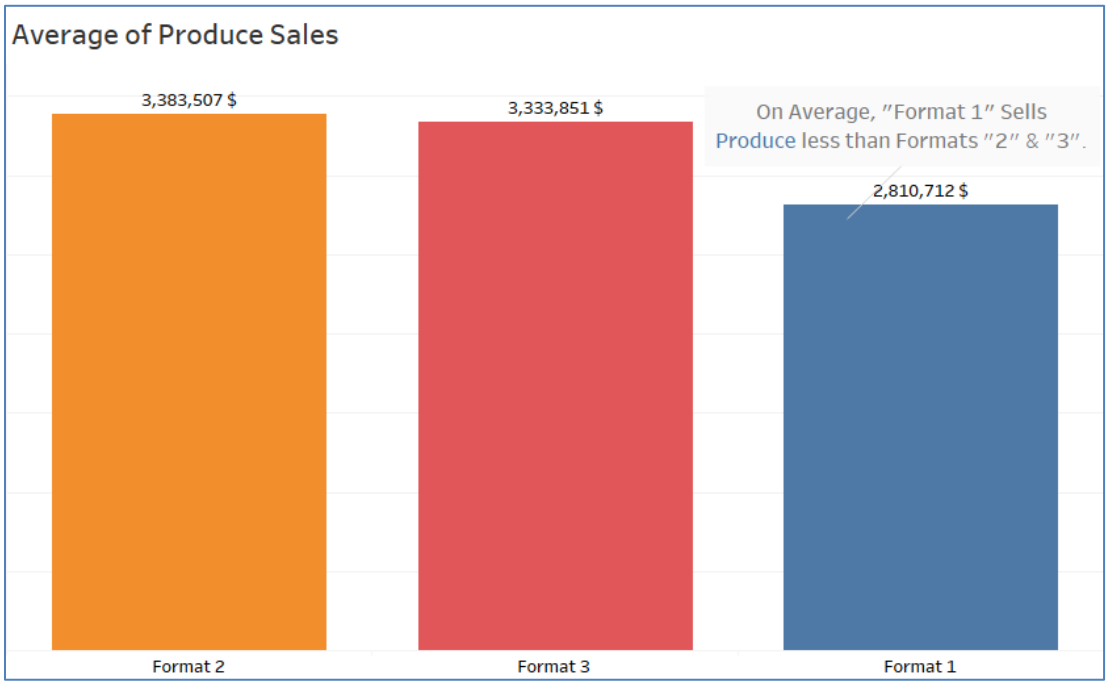
**(1)** <u>On Average</u>, "**Format 2**" Sells **Floral** **<u>more than</u>** Formats **"1" & "3"**.
While "**Format 1**" sells the **<u>smallest</u>** <u>amount</u> of **Floral**.



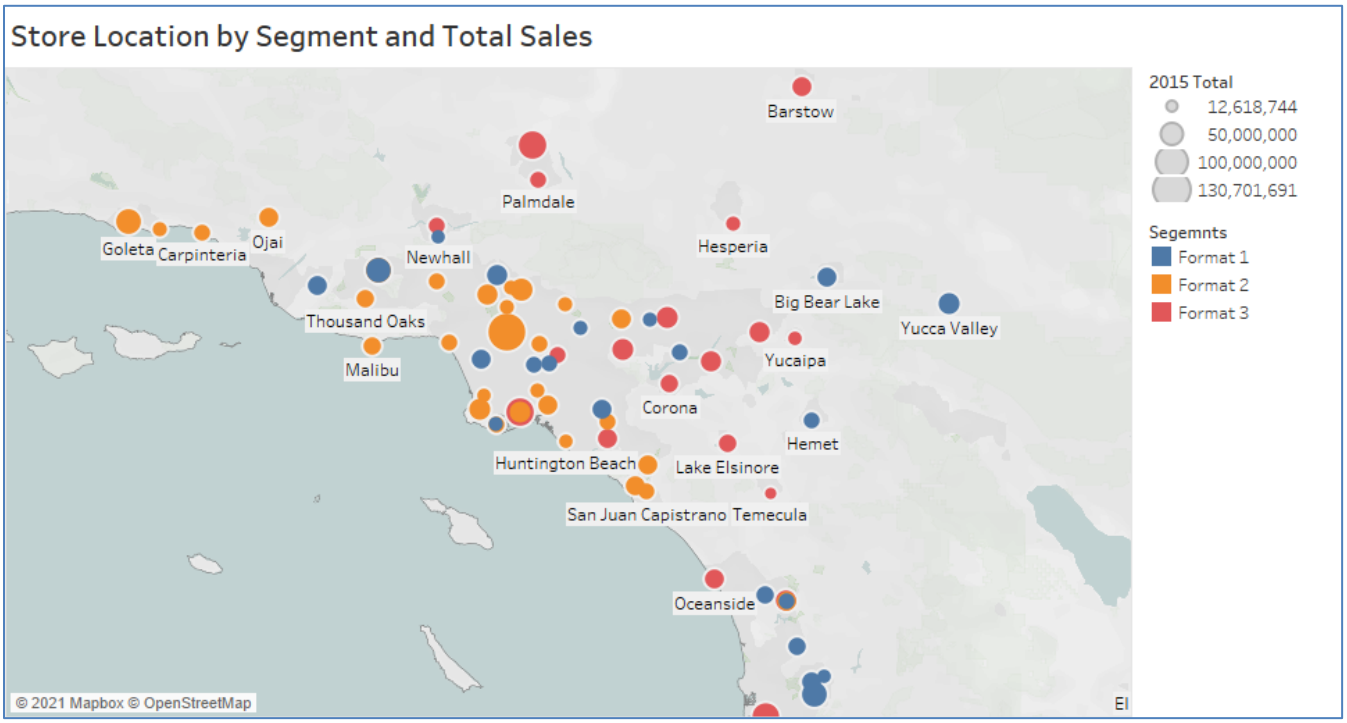

**(2)** <u>On Average</u>, "**Format 3**" Sells **General Merchandise** <u>more than</u> **Formats "1" & "2"**.



**(3)** On Average, "**Format 1**" Sells **Produce** less than **Formats "2" & "3"**.




4. Please provide a Tableau visualization (saved as a Tableau Public file) that shows the location of the stores, uses color to show cluster, and size to show total sales.



# Task 2: Formats for New Stores

1. What methodology did you use to predict the best store format for the new stores? Why did you choose that methodology? (Remember to Use a 20% validation sample with Random Seed = 3 to test differences in models.)

- The methodology I used to predict the best store format for the new stores is: **"Predictive Analysis" Methodology** using a **"Boosted Model"** for **"Non-Binary Classification Problem"**.

- I have chosen this methodology because: our **Target Variable** (*which is the Store Format*) is a **Categorical** Variable, with **3 values** (*which are Format 1, Format 2, and Format 3*).

- **I applied 3 classification models** (Decision Tree Model, Forest Model, and Boosted Model), **Evaluated** all of them against both "Estimation Sample" and "Validation Sample", and **Conclude** that the **Boosted Model** is the best one to use in predicting the best store format for the new stores.

- Using model validation (against validation dataset), The decision-making criteria are set based on:
  1. Best **Overall Accuracy** %.
  2. Best **F1** Score.
  3. Lowest **Error Rate** %
  4. Best **Precision** Average %.
  5. Best **Recall** Average %.

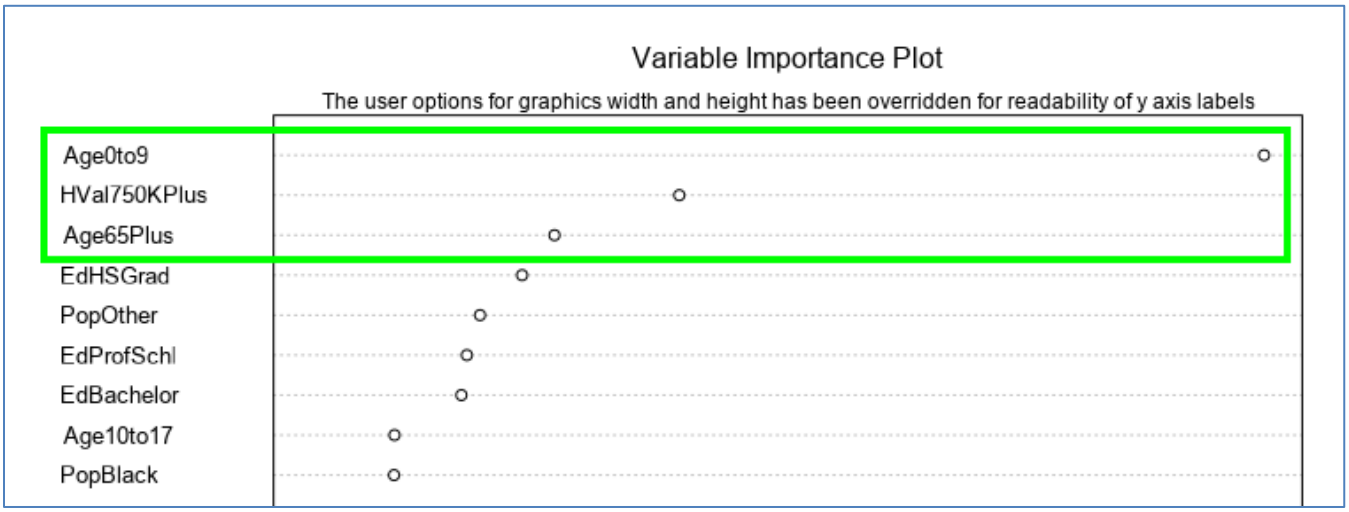
- Thus, the **Boosted Model** has been chosen because it **has the upper hand** over the **2 other models** in all above criteria, as demonstrated below:

| Measure | Decision Tree Model | Forest Model | Boosted Model |
|---|---|---|---|
| Overall **Accuracy** % | 65.0% | 71.0% | **76.0%** |
| **F1** Score % | 67.0% | 75.0% | **83.0%** |
| **Error** Rate Average % | 33.0% | 25.0% | **17.0%** |
| **Precision** Average % | 66.0% | 70.0% | **79.0%** |
| **Recall** Average % | 67.0% | 75.0% | **83.0%** |

| Model | Accuracy | F1 | Accuracy_Cluster 1 | Accuracy_Cluster 2 | Accuracy_Cluster 3 |
|---|---|---|---|---|---|
| DTree_Cluster | 0.6471 | 0.6667 | 0.5000 | 1.0000 | 0.5000 |
| Forest_Cluster | 0.7059 | 0.7500 | 0.5000 | 1.0000 | 0.7500 |
| Boosted_Cluster | 0.7647 | 0.8333 | 0.5000 | 1.0000 | 1.0000 |

- <u>The three most important variables are:</u>

  1) **Age0to9**
  2) **Hval750Kplus**
  3) **Age65Plus**




- *For **more details** about each model evaluation, please check pages numbers (**9**, **10**, & **11**).*

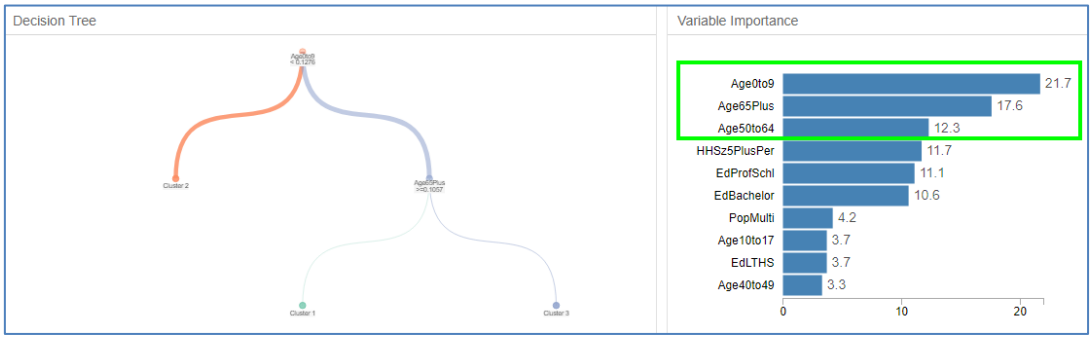
2. What do format each of the 10 new stores fall into? Please fill in the table below.

| Store Number | Segment |
|---|---|
| S0086 | Cluster **1** |
| S0087 | Cluster **2** |
| S0088 | Cluster **3** |
| S0089 | Cluster **2** |
| S0090 | Cluster **2** |
| S0091 | Cluster **3** |
| S0092 | Cluster **2** |
| S0093 | Cluster **3** |
| S0094 | Cluster **2** |
| S0095 | Cluster **2** |

# 1. Decision Tree Model

**Model Result**

**(Against Training Dataset)**




**Top 3 Important Predictor Variables** [Weight about **52%** from all predictors]

- Age0to9 **[21.7 %]**
- Age65Plus **[17.6 %]**
- Age50to64 **[12.3 %]**

**Model Validation Result**

**(Against Validation Dataset)**

| Model | Accuracy | F1 | Accuracy_Cluster 1 | Accuracy_Cluster 2 | Accuracy_Cluster 3 |
|---|---|---|---|---|---|
| DTree_Cluster | 0.6471 | 0.6667 | 0.5000 | 1.0000 | 0.5000 |
| Forest_Cluster | 0.7059 | 0.7500 | 0.5000 | 1.0000 | 0.7500 |
| Boosted_Cluster | 0.7647 | 0.8333 | 0.5000 | 1.0000 | 1.0000 |

**Confusion matrix of DTree_Cluster**

| | Actual_Cluster 1 | Actual_Cluster 2 | Actual_Cluster 3 |
|---|---|---|---|
| Predicted_Cluster 1 | 4 | 0 | 2 |
| Predicted_Cluster 2 | 3 | 5 | 0 |
| Predicted_Cluster 3 | 1 | 0 | 2 |

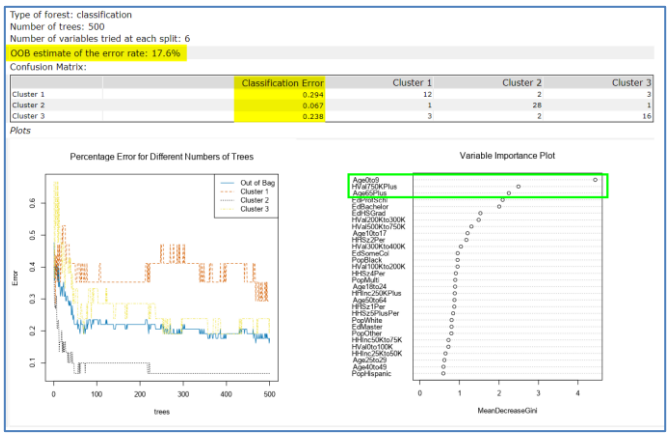
| Measure | Cluster 1 | Cluster 2 | Cluster 3 |
|---|---|---|---|
| **Precision** = (True / Class Predicted True Total) | **67.0%** | **63.0%** | **67.0%** |
| **Recall** = (True / Class Actual Total) | **50.0%** | **100.0%** | **50.0%** |
| **Error Rate** = (Rev. False / Class Actual Total) | **50.0%** | **00.0%** | **50.0%** |
| **Overall Accuracy** = (Predicted True Total / n) | (4 + 5 + 2) / 17 = **65.0%** | | |
| **Null Accuracy** = (Dominated Class Actual Total / n) | (4 + 3 + 1) / 17 = **47.0%** | | |

- **Overall** Accuracy: **65.0%**
- **Error Rate** Average: **33.0%**
- **Precision** Average: **66.0%**
- **Recall** Average: **67.0%**

## 2. Forest Model

**Model Result**

**(Against Training Dataset)**




**Top 3 Important Predictor Variables**

- Age0to9
- Hval750Kplus
- Age65Plus

This model has **high** error rate at prediction [**Error Rate: 17.6%**], especially for **Cluster 1** by **29.0%**.

**Model Validation Result**

**(Against Validation Dataset)**

| Model | Accuracy | F1 | Accuracy_Cluster 1 | Accuracy_Cluster 2 | Accuracy_Cluster 3 |
|---|---|---|---|---|---|
| DTree_Cluster | 0.6471 | 0.6667 | 0.5000 | 1.0000 | 0.5000 |
| Forest_Cluster | 0.7059 | 0.7500 | 0.5000 | 1.0000 | 0.7500 |
| Boosted_Cluster | 0.7647 | 0.8333 | 0.5000 | 1.0000 | 1.0000 |

**Confusion matrix of Forest_Cluster**

| | Actual_Cluster 1 | Actual_Cluster 2 | Actual_Cluster 3 |
|---|---|---|---|
| Predicted_Cluster 1 | 4 | 0 | 1 |
| Predicted_Cluster 2 | 2 | 5 | 0 |
| Predicted_Cluster 3 | 2 | 0 | 3 |

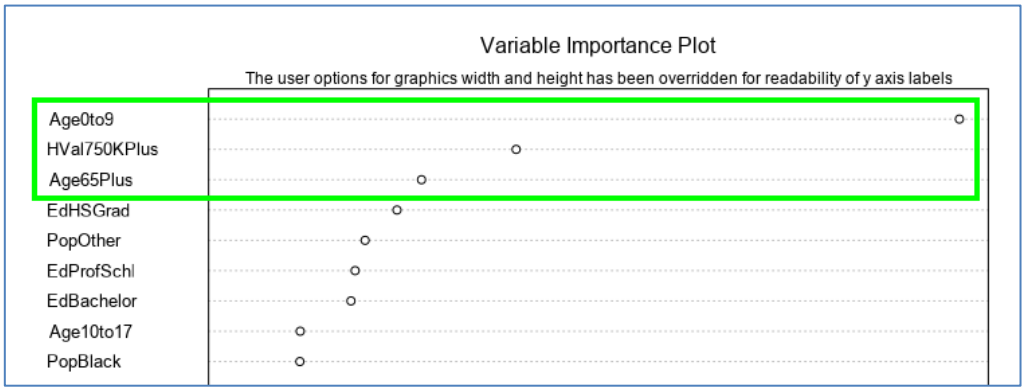
| Measure | Cluster 1 | Cluster 2 | Cluster 3 |
|---|---|---|---|
| **Precision** = (True / Class Predicted True Total) | **80.0%** | **71.0%** | **60.0%** |
| **Recall** = (True / Class Actual Total) | **50.0%** | **100.0%** | **75.0%** |
| **Error Rate** = (Rev. False / Class Actual Total) | **50.0%** | **00.0%** | **25.0%** |
| **Overall Accuracy** = (Predicted True Total / n) | (4 + 5 + 3) / 17 = **71.0%** | | |
| **Null Accuracy** = (Dominated Class Actual Total / n) | (4 + 2 + 2) / 17 = **47.0%** | | |

- **Overall** Accuracy: **71.0%**
- **Error Rate** Average: **25.0%**
- **Precision** Average: **70.0%**
- **Recall** Average: **75.0%**

## 3. Boosted Model

**Model Result**

**(Against Training Dataset)**




**Top 3 Important Predictor Variables**

- Age0to9
- Hval750Kplus
- Age65Plus

**Model Validation Result**

**(Against Validation Dataset)**

| Model | Accuracy | F1 | Accuracy_Cluster 1 | Accuracy_Cluster 2 | Accuracy_Cluster 3 |
|---|---|---|---|---|---|
| DTree_Cluster | 0.6471 | 0.6667 | 0.5000 | 1.0000 | 0.5000 |
| Forest_Cluster | 0.7059 | 0.7500 | 0.5000 | 1.0000 | 0.7500 |
| Boosted_Cluster | 0.7647 | 0.8333 | 0.5000 | 1.0000 | 1.0000 |

**Confusion matrix of Boosted_Cluster**

| | Actual_Cluster 1 | Actual_Cluster 2 | Actual_Cluster 3 |
|---|---|---|---|
| Predicted_Cluster 1 | 4 | 0 | 0 |
| Predicted_Cluster 2 | 2 | 5 | 0 |
| Predicted_Cluster 3 | 2 | 0 | 4 |

| Measure | Cluster 1 | Cluster 2 | Cluster 3 |
|---|---|---|---|
| **Precision** = (True / Class Predicted True Total) | **100.0%** | **71.0%** | **67.0%** |
| **Recall** = (True / Class Actual Total) | **50.0%** | **100.0%** | **100.0%** |
| **Error Rate** = (Rev. False / Class Actual Total) | **50.0%** | **00.0%** | **00.0%** |
| **Overall Accuracy** = (Predicted True Total / n) | (4 + 5 + 4) / 17 = **76.0%** | | |
| **Null Accuracy** = (Dominated Class Actual Total / n) | (4 + 2 + 2) / 17 = **47.0%** | | |

- **Overall** Accuracy: **76.0%**
- **Error Rate** Average: **17.0%**
- **Precision** Average: **79.0%**
- **Recall** Average: **83.0%**

# <u>Task 3</u>: Predicting Produce Sales

1. What type of ETS or ARIMA model did you use for each forecast? Use ETS (A,M,N) or ARIMA (AR, I, MA) notation. How did you come to that decision?

- <u>The type I used for each forecast is</u>: **ETS(MNM)**.

- <u>I came to that decision after took the below steps:</u>

  1. I **prepared the data** to match Time Series Criteria.
  2. I determined the **Error, Trend, and Seasonality** using Time Series & Decomposition Plots
  3. I filtered out the last 6 Records, as a **Holdout Sample**.
  4. I created **3 ETS Models** and **3 ARIMA** Models, in sake for comparison.
  5. The **ETS(MNM)** has the **best forecasting abilities** and the **least Errors** in prediction.
  6. Thus, I used it for forecasting existing and new stores.

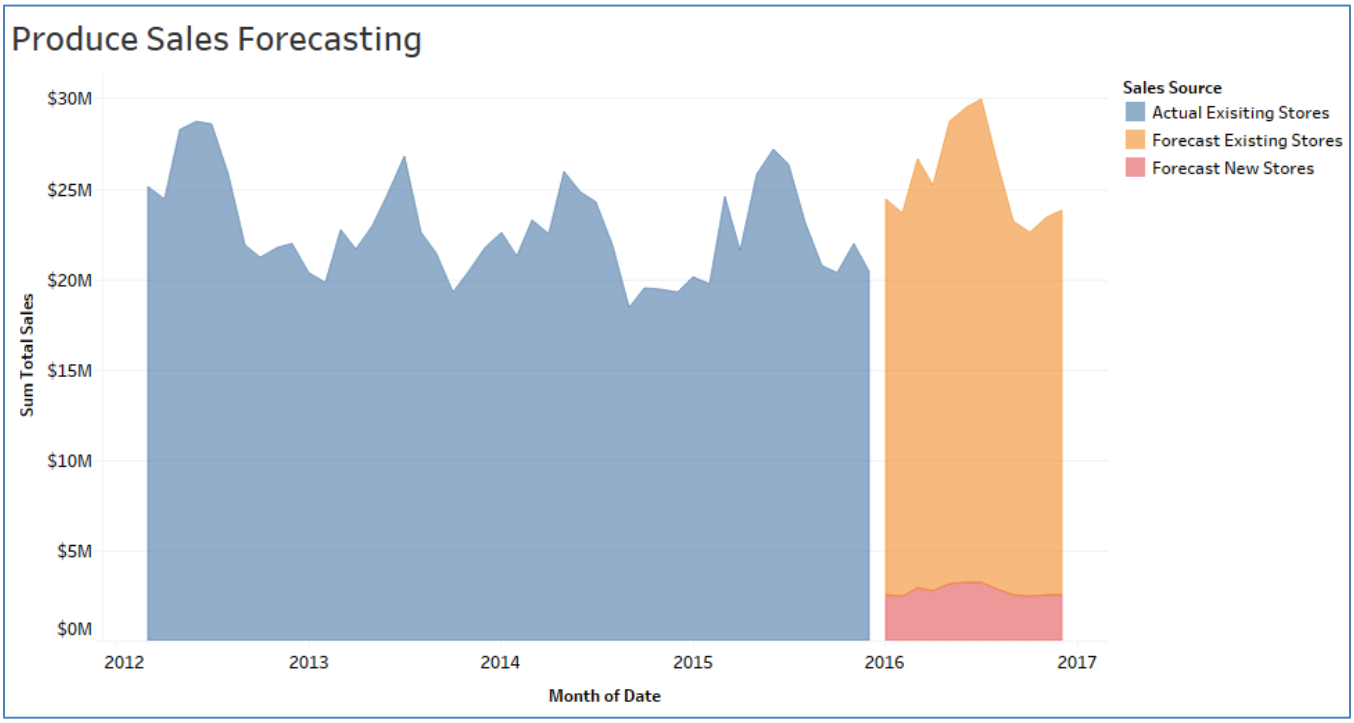
- *For **more details** about every step taken, please check pages numbers (from **14** to **31**).*

2. Please provide a table of your forecasts for existing and new stores. Also, provide visualization of your forecasts that includes historical data, existing stores forecasts, and new stores forecasts.

- **The Table:**

| Month | Forecast New Stores | Forecast Existing Stores |
|---|---|---|
| Jan-2016 | 2,563,357.9 | 21,829,060 |
| Feb-2016 | 2,483,924.7 | 21,146,329.6 |
| Mar-2016 | 2,910,944.1 | 23,735,686.9 |
| Apr-2016 | 2,764,881.9 | 22,409,515.3 |
| May-2016 | 3,141,305.9 | 25,621,828.7 |
| Jun-2016 | 3,195,054.2 | 26,307,858 |
| Jul-2016 | 3,212,391 | 26,705,092.6 |
| Aug-2016 | 2,852,385.8 | 23,440,761.3 |
| Sep-2016 | 2,521,697.2 | 20,640,047.3 |
| Oct-2016 | 2,466,750.9 | 20,086,270.5 |
| Nov-2016 | 2,557,744.6 | 20,858,120 |
| Dec-2016 | 2,530,510.8 | 21,255,190.2 |

- **The Visual:**


Produce Sales Forecasting
$30M
$25M
$20M
$15M
$10M
$5M
$0M
Sum Total Sales
2012
2013
2014
2015
2016
2017
Month of Date
Sales Source
Actual Exisiting Stores
Forecast Existing Stores
Forecast New Stores

# Step 1: Analysis Plan

*Looking at data set and determine whether the data is appropriate to use time series models. Determining which records should be held for validation later.*

## *Existing Stores*

### Data Processing

- We are trying to get the "**Monthly Produce Total Sales**" for **existing** stores before forecasting.
- The data we have is at **daily store level**, so some data transformation/processing is required to convert it to be at **monthly level per year only**.

| Record | Store | Day | Month | Year | Produce | Cluster |
|---|---|---|---|---|---|---|
| 1 | S0001 | 01 | 03 | 2012 | 4,542.28 | 1 |
| 2 | S0001 | 02 | 03 | 2012 | 9,503.45 | 1 |
| 3 | S0001 | 03 | 03 | 2012 | 12,793.92 | 1 |
| 4 | S0001 | 04 | 03 | 2012 | 16,635.77 | 1 |

- We can accomplish this using a summarize tool as below:
  - Group by Year.
  - Group by Month.
  - Sum Produce.

| Record | Year | Month | Sum_Produce |
|---|---|---|---|
| 1 | 2012 | 03 | 25,151,525.84 |
| 2 | 2012 | 04 | 24,406,048.39 |
| 3 | 2012 | 05 | 28,249,539.01 |
| 4 | 2012 | 06 | 28,691,364.32 |

- Now, the dataset at our hand is a "**Monthly Produce Total Sales**" data with **No missing values**.
- It also **meets the criteria** of a time series dataset, as it is:
  - It is **time dependent** in a **continuous order** "continuous time monthly intervals".
  - Sales values exist as a **measure** at **sequential** monthly order with **equal spacing**.
  - Each time unit (month) has **at most 1 data point** (which is the monthly **Sum_Produce**).

### Holdout Sample

- For the holdout sample (validation sample), it's good to use the **same number of periods** that we'll be forecasting for. So, it is **preferred** to use **12 Records** as the **holdout** sample.

- However, **we do not have that much data**, so using a 12 months holdout would remove too much of the data. So, we'll use a **6 month** holdout sample for the **TS Compare tool**.
- I have filtered out the last 6 records, **2015-07** to **2015-12**, as a holdout sample so that I can check the accuracy of my model to forecast predicted values against the actual values.

## *New Stores*

- We are trying to get "**Monthly Produce Average Total Sales**" per **cluster** for **existing** stores before forecasting.
- The data we have is at **daily store level**, so some data transformation/processing is required to convert it to be at **monthly level per year for each cluster alone**.

| Record | Store | Day | Month | Year | Produce | Cluster |
|---|---|---|---|---|---|---|
| 1 | S0001 | 01 | 03 | 2012 | 4,542.28 | 1 |
| 2 | S0001 | 02 | 03 | 2012 | 9,503.45 | 1 |
| 3 | S0001 | 03 | 03 | 2012 | 12,793.92 | 1 |
| 4 | S0001 | 04 | 03 | 2012 | 16,635.77 | 1 |

- We can accomplish this using a 2 summarize tools tool as below:
  - Group by Store.
  - Group by Cluster.
  - Group by Year.
  - Group by Month.
  - Sum Produce.

| Record | Store | Cluster | Year | Month | Sum_Produce |
|---|---|---|---|---|---|
| 1 | S0001 | 1 | 2012 | 03 | 225,673.98 |
| 2 | S0001 | 1 | 2012 | 04 | 209,257.48 |
| 3 | S0001 | 1 | 2012 | 05 | 239,711.93 |
| 4 | S0001 | 1 | 2012 | 06 | 249,080.61 |

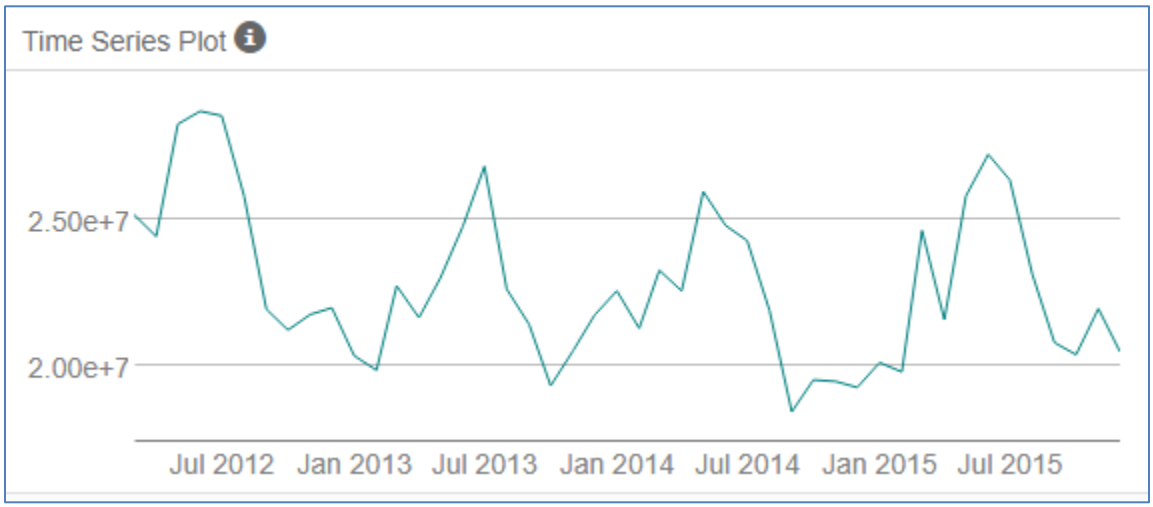
- Then:
  - Group by Cluster.
  - Group by Year.
  - Group by Month.
  - Avg Sum_Produce.

| Record | Cluster | Year | Month | Avg_Sum_Produce |
|---|---|---|---|---|
| 1 | 1 | 2012 | 03 | 291,606.651667 |
| 2 | 1 | 2012 | 04 | 279,029.90375 |
| 3 | 1 | 2012 | 05 | 315,221.33875 |
| 4 | 1 | 2012 | 06 | 323,170.302917 |

# Step 2: Determining Trend, Seasonal, and Error components

*By looking at **Time Series Graph**, and **Decomposition Plot** determine the **Trend**, **Seasonal**, and **Error** components.*

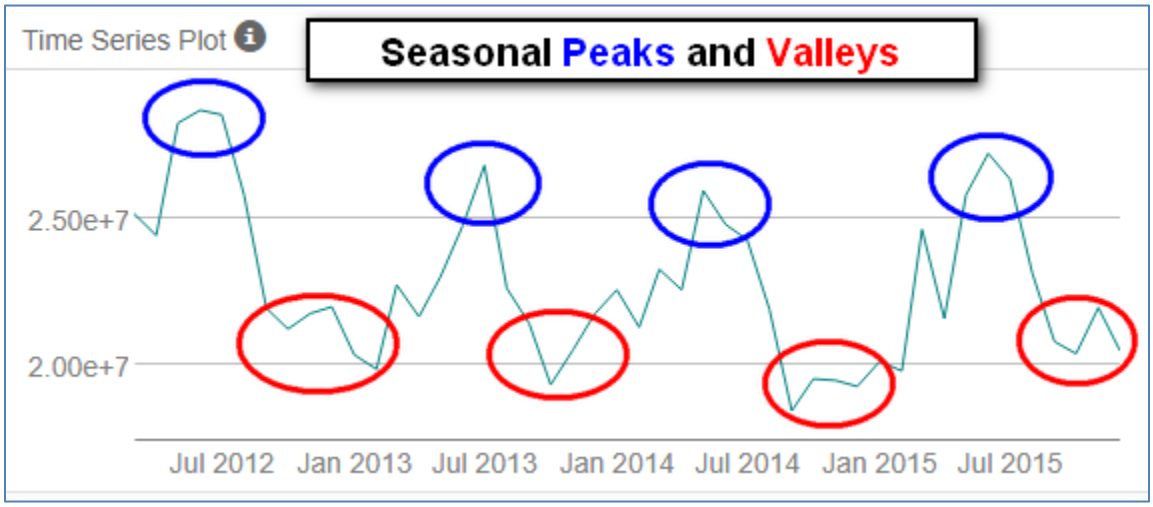
## *Time Series Graph*




*By looking at **Time Series Graph**, we can notice the below…*

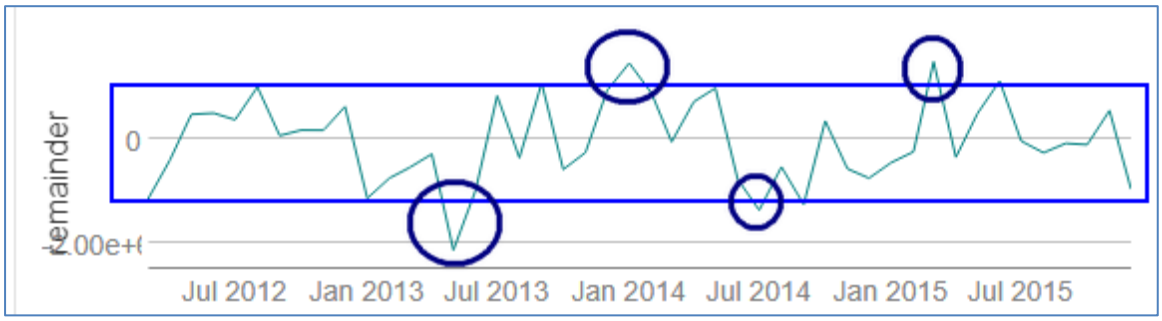
- There is no clear Upward nor Downward trend at the data., it seems to have a "**Stationary or Horizontal Trend**", which means "**No Trend**".
- We will make sure from our notice using **Trend Plot** at **Decomposition** Plot.

- It is a **seasonal data**, as we can see at **Jul peaks** and **Sep** and **Oct Valleys**.
- We will make sure from our notice using **Seasonal Plot** at **Decomposition** Plot.




## *Decomposition Plot*

### Error

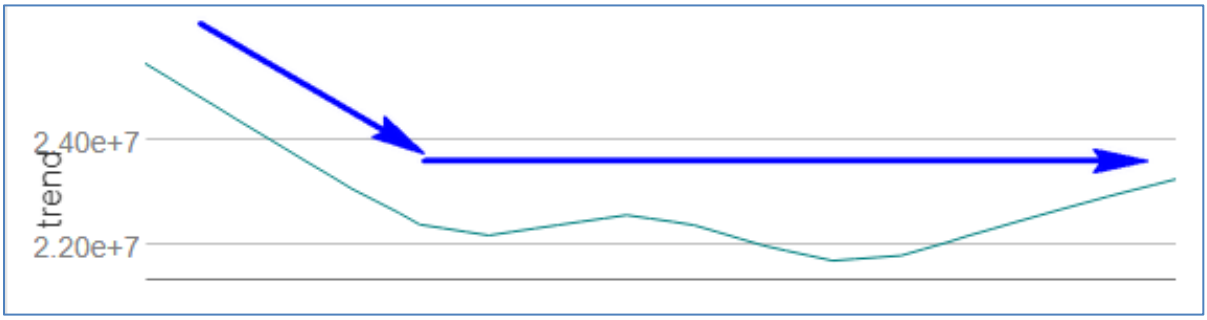



- <u>As we know</u>,
  - If the error has **constant variance** over time (**peaks** and **valleys** are about the **same size**), then we apply it **additively**.
  - If the error is **growing** or **shrinking** over time, then we would apply the error **multiplicatively**.

- <u>In this case as we see</u>, the error is **growing** over time, then we would apply the error **Multiplicatively (M)**.

- <u>By the way</u>, the ETS tool **Auto** selects **Multiplicative** also.

## Trend:

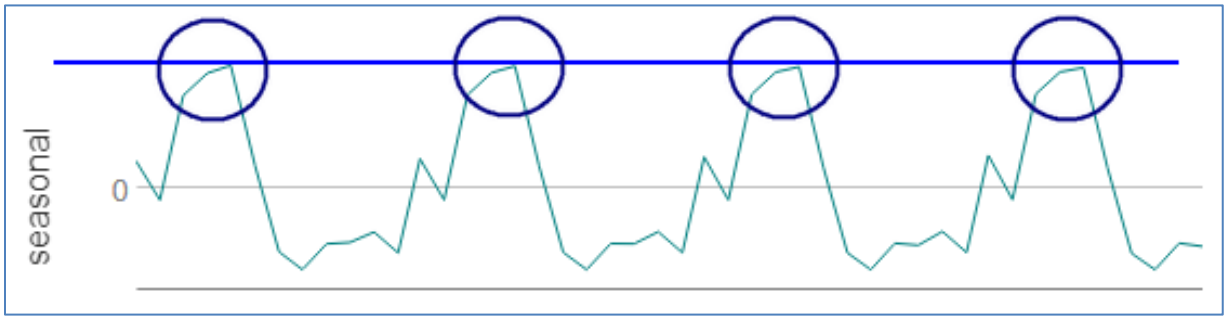



- As we know,
  - If the trend is **linear**, then we apply it **additively**.
  - If the trend line **grows** or **shrinks exponentially**, then we would apply the trend **multiplicatively**.

- In this case as we see, Although the trend seems going **down at the beginning**, but generally we can see it goes **horizontally most of the time**, so, there is "**No Trend**", and we should **not** apply the trend **Additively (A) nor Multiplicatively (M)**.
  So, we would apply **No Trend (N)**.

- By the way, the ETS tool **Auto** selects **None** also.

## Seasonality



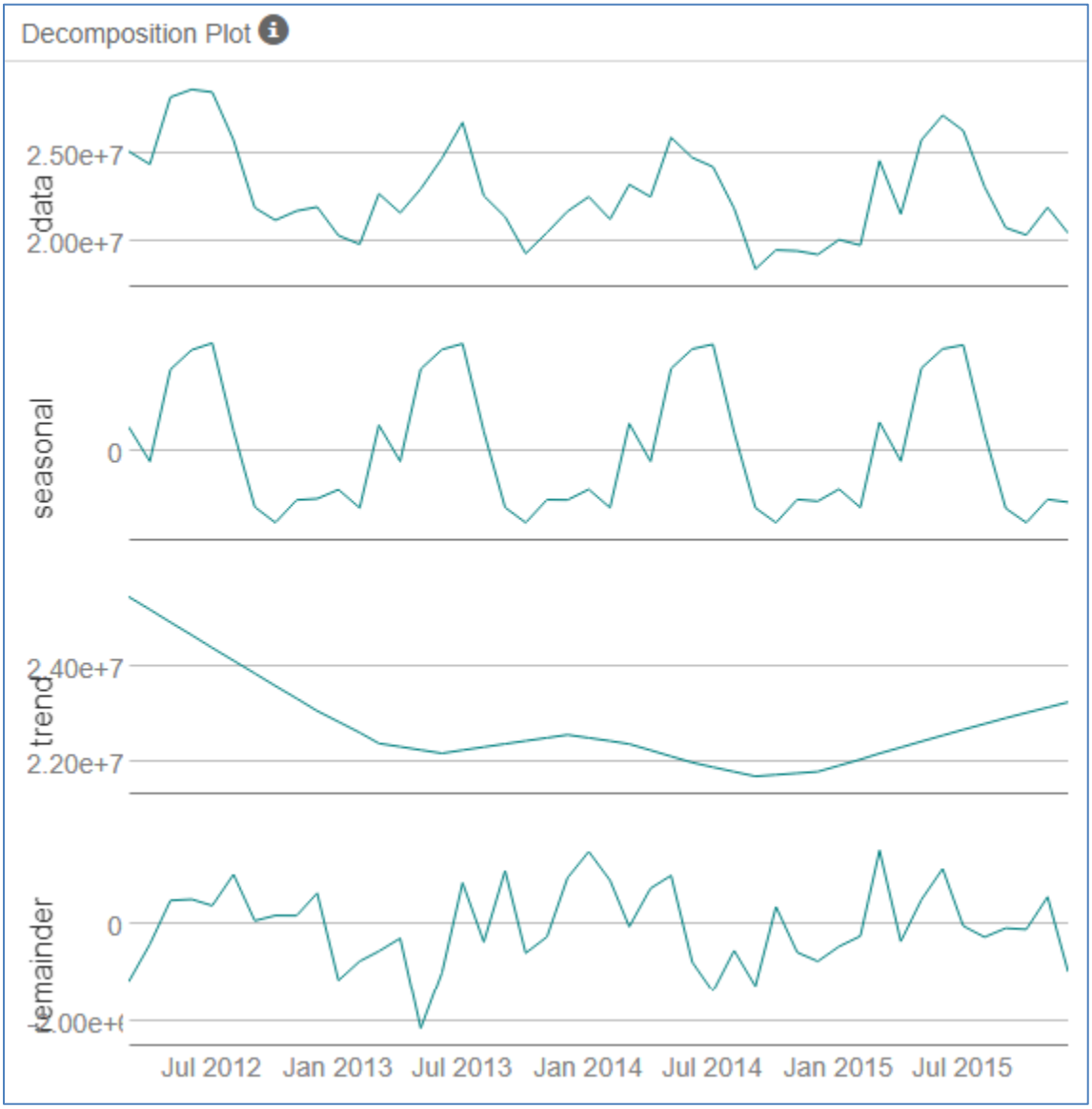

- As we know,
  - If the **peaks** and **valleys** for seasonality are **constant** over time, then we apply it **additively**.
  - the seasonality **grows** or **shrinks** overtime, then we would apply the season **multiplicatively**.
  - *We don't have to consider if the change is linear or exponential*, **only** if it is growing or shrinking over time.

- In this case as we see, Seasonality is **shrinking slightly over time** (the **peaks** are **decreasing**, just very **slowly**), so we would apply it **Multiplicatively (M)**.

- By the way, the ETS tool **Auto** selects **Multiplicative** also.

# Step 3: Building Time Series Models

As we saw above, decomposition tree plot gives us the sense about the appropriate measurements to apply to **ARIMA** and **ETS** models.



# *ETS Model*

## Model Configuration

- From the decomposition plot we can obtain the necessary information to define our terms for the **ETS** model.
  - **<u>The Error</u>:** ***changes in magnitude*** as the series goes along so a **multiplicative (M)** method will be used.
  - **<u>The Trend</u>:** line exhibits **No Trend behavior** so we will use an **None (N)** method.
  - **<u>The Seasonality</u>:** changes in **magnitude** each year so a **multiplicative (M)** method is necessary
- This leaves us with an **ETS(M, N, M)** model.
- <u>For a **comparison** purpose and for **checking** our readings</u>, we could also perform:
  - **ETS(M, $A_d$, M)** model with **Dampened** Parameter.
  - **ETS(Auto)** model.
- **ETS(M, N, M)** is the **best** **ETS** model to be chosen based on below **evaluation metrices**.
- It also the same model suggested by **ETS(Auto)** model.

## In-sample errors

- The in-sample error measures give us a look at **how well** our model can predict future.

**ETS(M, N, M)** model

Method:
ETS(M,N,M)

In-sample error measures:

| ME | RMSE | MAE | MPE | MAPE | MASE | ACF1 |
|---|---|---|---|---|---|---|
| 3502.9443415 | 969051.6076376 | 787577.7006835 | -0.1381187 | 3.4677635 | 0.4396486 | 0.0077488 |

Information criteria:

| AIC | AICc | BIC |
|---|---|---|
| 1279.4203 | 1299.4203 | 1304.7535 |

**ETS(M, $A_d$, M)** model with **Dampened** Parameter

Method:
ETS(M,Ad,M)

In-sample error measures:

| ME | RMSE | MAE | MPE | MAPE | MASE | ACF1 |
|---|---|---|---|---|---|---|
| 70427.8970905 | 961384.5898397 | 788678.3905704 | 0.179046 | 3.4496582 | 0.440263 | -0.0436975 |

Information criteria:

| AIC | AICc | BIC |
|---|---|---|
| 1285.2405 | 1317.8119 | 1315.6403 |

**ETS_Auto(M, N, M)** model.

Method:
ETS(M,N,M)

In-sample error measures:

| ME | RMSE | MAE | MPE | MAPE | MASE | ACF1 |
|---|---|---|---|---|---|---|
| 3502.9443415 | 969051.6076376 | 787577.7006835 | -0.1381187 | 3.4677635 | 0.4396486 | 0.0077488 |

Information criteria:

| AIC | AICc | BIC |
|---|---|---|
| 1279.4203 | 1299.4203 | 1304.7535 |

- For ETS**(M, N, M)**: 3 key components to look at are the **RMSE**, which shows the in-sample **standard deviation**, and the **MASE** which can be used to compare **forecasts of different models**, and **AIC**, which can be used for **intra-model** comparison (Model Configuration)
    - At **RMSE**, we can see that our variance is about **969,000** units around the mean.
    - The **MASE** shows a fairly strong forecast at **0.44** with its value falling well below the generic **1.00**, the commonly accepted MASE threshold for model accuracy.
    - **AIC** value **1279**, which is the best AIC value, comparing with the other **ETS** model.

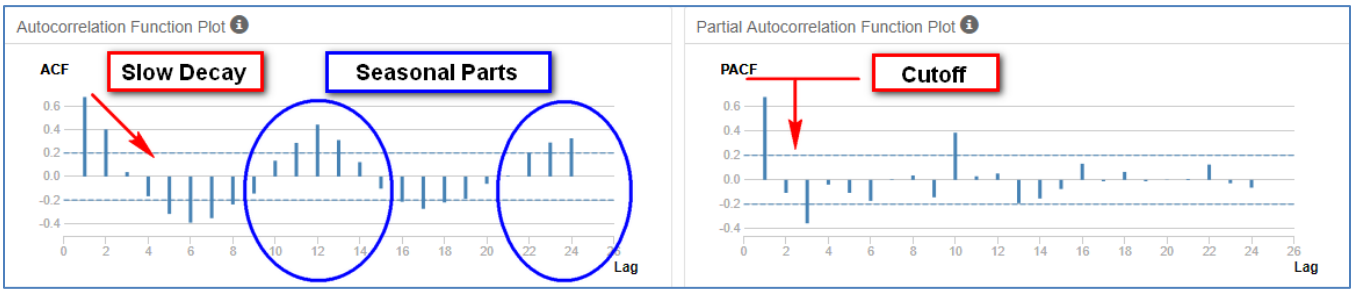
# *ARIMA Model*

## Model Configuration

- Since there are **seasonal components** found in the time series, we will use an :
    - **ARIMA (p, d, q) (P, D, Q)m model for forecasting.**

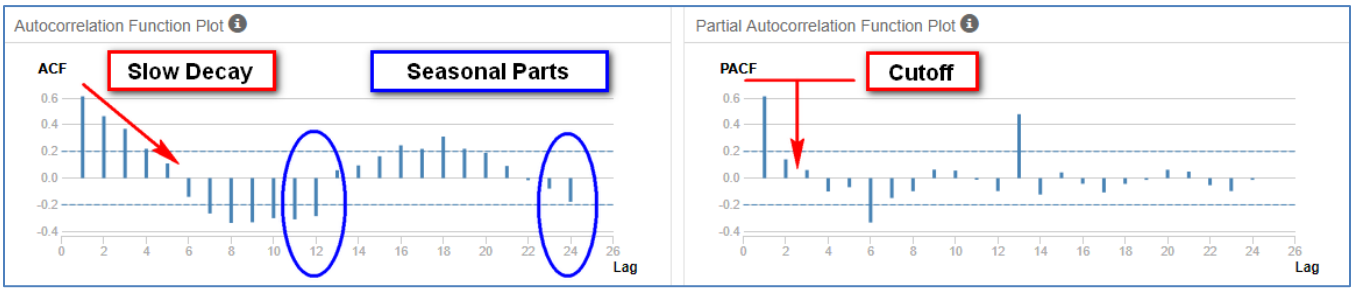
**Time Series ACF and PACF:**

- The **ACF** presents **slowly decaying** serial correlations towards **0** with increases at the seasonal lags. Since serial correlation is high, we will need to **seasonally difference** the series.

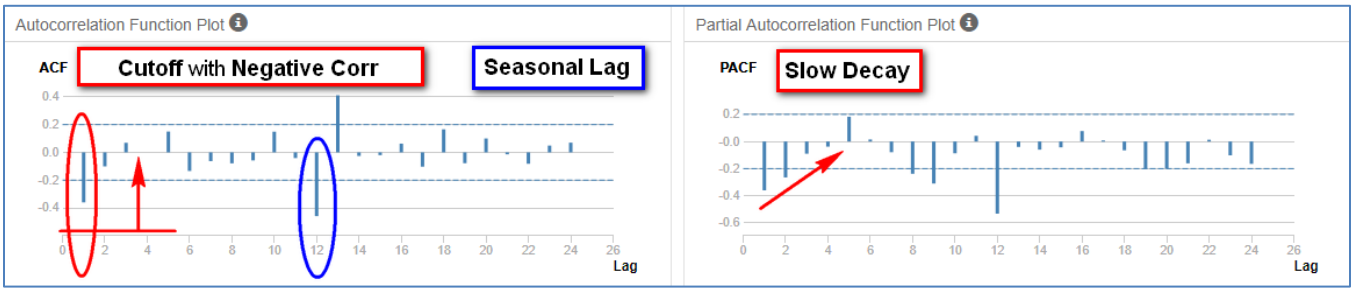



**Seasonal Difference ACF and PACF:**

- The seasonal difference presents **similar ACF and PACF results** as the initial plots without differencing, only **slightly less correlated**. In order to remove correlation, **we will need to difference further**.



**Seasonal First Difference:**




- The seasonal first difference of the series has **removed most of the significant lags** from the **ACF and PACF** so there is **no need for further differencing**.

- The remaining correlation can be accounted for using **autoregressive** and **moving average** terms.

- We did 2 differencing, 1 for the Non-Seasonal part (d) and 1 for the Seasonal part (D), so:
    - The **differencing** terms will be **d(1)** and **D(1).**

- The ACF plot shows **Cutoff towards 0**, a **strong negative correlation** at **lag 1** which is confirmed in the **PACF**, as we see **slow decay towards 0**.
    - This suggests an **MA(1) model** since there is **only 1 significant lag**.

- The **seasonal lags** (lag 12, 24, etc.) has a **significant correlation** in the **ACF**.
    - This suggests an **MA(1) model** at seasonal part since there is **only 1 significant lag**.

- Therefore, the model terms for my ARIMA model are:
    - **ARIMA(0, 1, 1)(0, 1, 1)[12]**

- For a **comparison** purpose and for **checking** our readings, we could also perform:
    - **ARIMA(0, 1, 1)(0, 2, 1)[12]**
    - **ARIMA (Auto)** model.

- **ARIMA(0, 1, 1)(0, 1, 1)[12]** is the **best ARIMA** model to be chosen based on below **evaluation metrices**.

- Although It is not the same model suggested by **ARIMA(Auto)** model.

## In-sample errors

- The in-sample error measures give us a look at **how well** our model can predict future.

**ARIMA(0, 1, 1)(0, 1, 1)[12]**

Information Criteria:

| AIC | AICc | BIC |
|---|---|---|
| 849.8292 | 850.8727 | 853.7167 |

In-sample error measures:

| ME | RMSE | MAE | MPE | MAPE | MASE | ACF1 |
|---|---|---|---|---|---|---|
| 150815.8641194 | 935292.1712234 | 628801.7029024 | 0.6312352 | 2.7761535 | 0.3510153 | -0.0469226 |

**ARIMA(0, 1, 1)(0, 2, 1)[12]**

Information Criteria:

| AIC | AICc | BIC |
|---|---|---|
| 504.5324 | 506.7142 | 506.6565 |

In-sample error measures:

| ME | RMSE | MAE | MPE | MAPE | MASE | ACF1 |
|---|---|---|---|---|---|---|
| -187514.1474174 | 1887887.7875214 | 946778.7504852 | -1.1642423 | 4.2307231 | 0.5285192 | -0.0073192 |

**ARIMA_Auto(1, 0, 0)(1, 1, 0)[12]**

Information Criteria:

| AIC | AICc | BIC |
|---|---|---|
| 880.4445 | 881.4445 | 884.4411 |

In-sample error measures:

| ME | RMSE | MAE | MPE | MAPE | MASE | ACF1 |
|---|---|---|---|---|---|---|
| -102530.8325034 | 1042209.8528363 | 738087.5530941 | -0.5465069 | 3.3006311 | 0.4120218 | -0.1854462 |

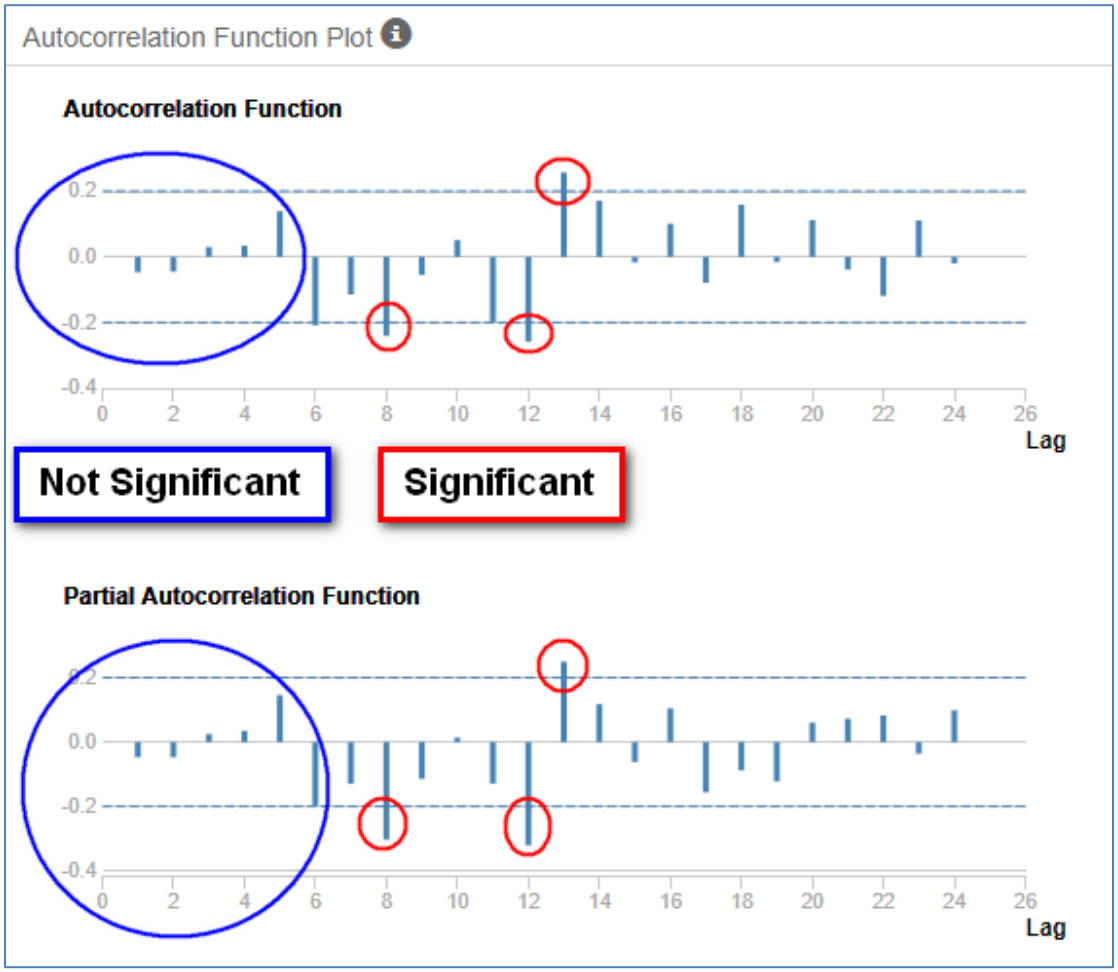
- For **ARIMA(0, 1, 1)(0, 1, 1)[12]** model: 3 key components to look at are the **RMSE**, which shows the in-sample **standard deviation**, and the **MASE** which can be used to compare **forecasts of different models**, and **AIC**, which can be used for **intra-model** comparison (Model Configuration)

    - At **RMSE**, we can see that our variance is about **935,000** units around the mean.

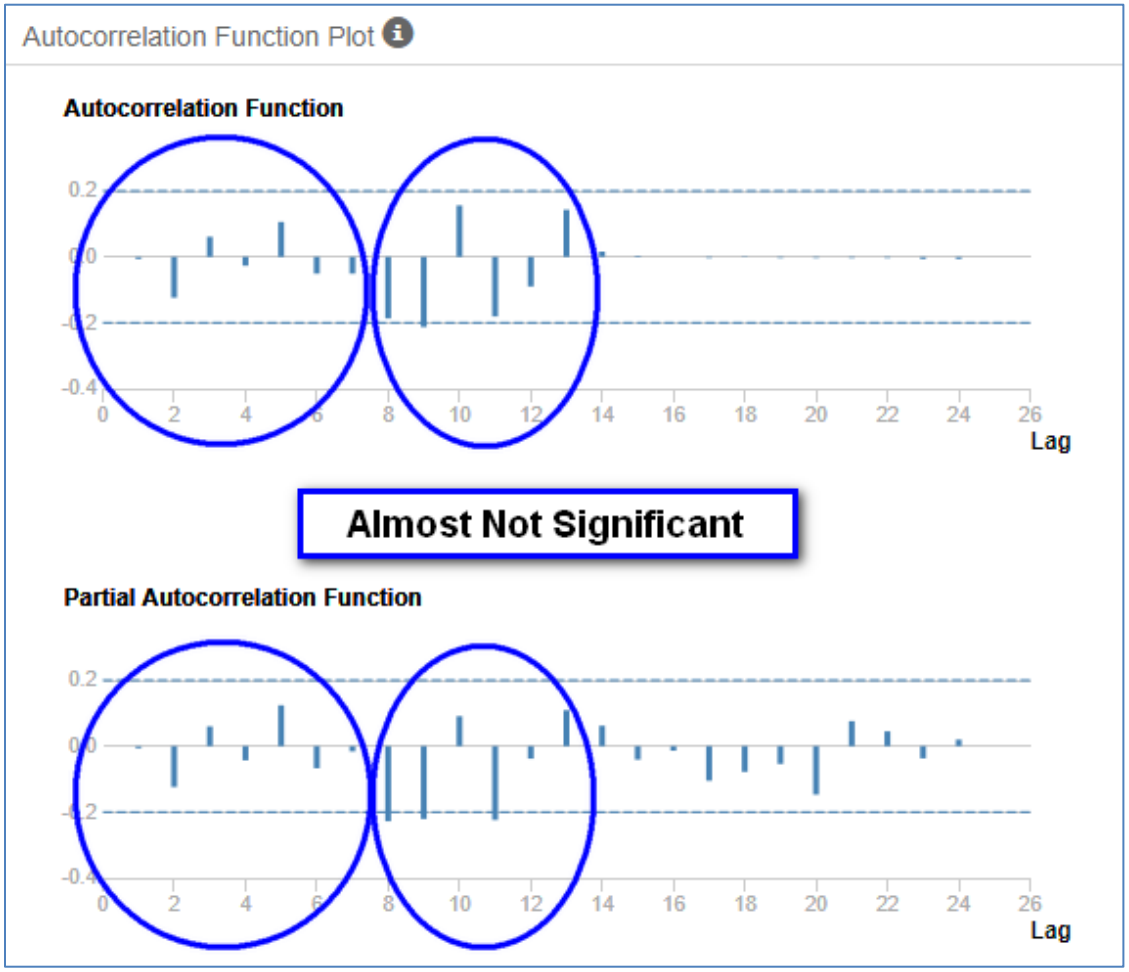
    - The **MASE** shows a fairly strong forecast at **0.35** with its value falling well below the generic **1.00**, the commonly accepted MASE threshold for model accuracy.

    - **AIC** value **849** is better than **ARIMA_Auto model**.

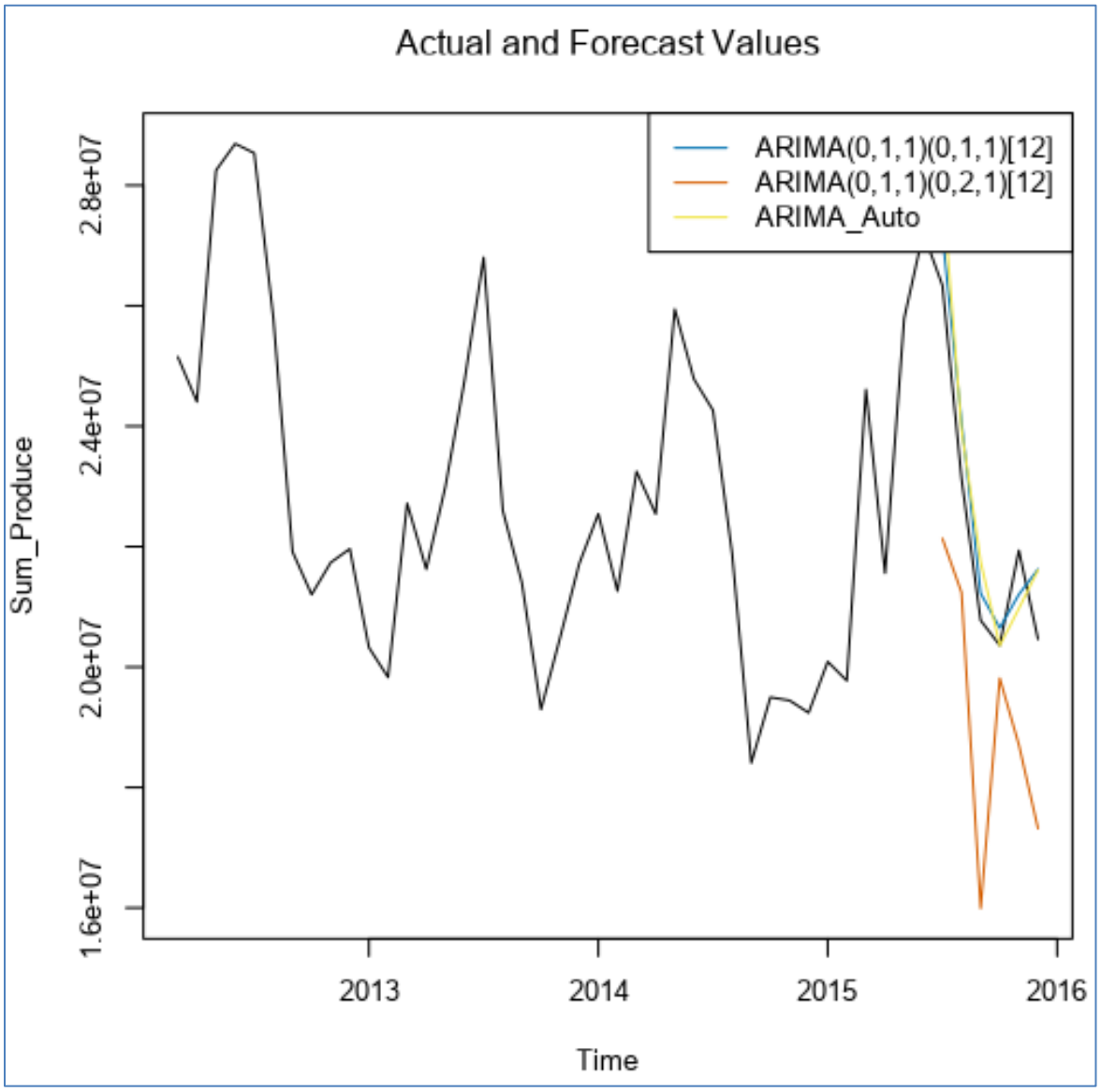
- The **ACF** and **PACF** results for the **ARIMA(0, 1, 1)(0, 1, 1)[12]** model shows that:
    - **No** significantly correlated **Non-Seasonal lags** suggesting **no** need for adding additional AR() or MA() terms.
    - **Little** significantly correlated **Seasonal lags** suggesting **a need for** adding additional AR() or MA() terms.



- Modifying ARIMA model to be **ARIMA(0, 1, 1)(0, 2, 1)[12]** instead of **ARIMA(0, 1, 1)(0, 1, 1)[12]**, for considering the **Significant Seasonal lags**, <u>results</u>:
    - A better **AIC** Value **504**, versus **849.**
    - **No significantly** correlated **Seasonal** nor **Non-Seasonal lags** suggesting **no** need for adding additional AR() or MA() terms.




- **<u>However</u>,** that will the model vulnerable to **Overfitting** issue, as it clear at <u>below graph</u> from the holdout sample result.


Actual and Forecast Values
ARIMA(0,1,1)(0,1,1)[12]
ARIMA(0,1,1)(0,2,1)[12]
ARIMA_Auto
Sum_Produce
1.6e+07
2.0e+07
2.4e+07
2.8e+07
2013
2014
2015
2016
Time

# Step 4: Forecast

- When fitting a forecasting model, we can use a **series of identifiers** that help us choose the best model.

## In-Sample error measures

- When comparing the two in-sample error measures we used, the **RMSE** and **MASE**, we see **very similar results**.
- The ARIMA model does have a **narrower standard deviation** but **only by a few** thousand units.
- Further investigation shows that the **MAPE** and **ME** of the **ARIMA** model are **lower** than the **ETS**. This suggests that, on average, the **ARIMA model** misses its forecast by a **lesser amount**.
- However, the **final decision** will be based on, the **Holdout Sample** error measures.

**ETS**

Method:
ETS(M,N,M)

In-sample error measures:

| ME | RMSE | MAE | MPE | MAPE | MASE | ACF1 |
|---|---|---|---|---|---|---|
| 3502.9443415 | 969051.6076376 | 787577.7006835 | -0.1381187 | 3.4677635 | 0.4396486 | 0.0077488 |

**ARIMA**

Method: ARIMA(0,1,1)(0,1,1)[12]

In-sample error measures:

| ME | RMSE | MAE | MPE | MAPE | MASE | ACF1 |
|---|---|---|---|---|---|---|
| 150815.8641194 | 935292.1712234 | 628801.7029024 | 0.6312352 | 2.7761535 | 0.3510153 | -0.0469226 |

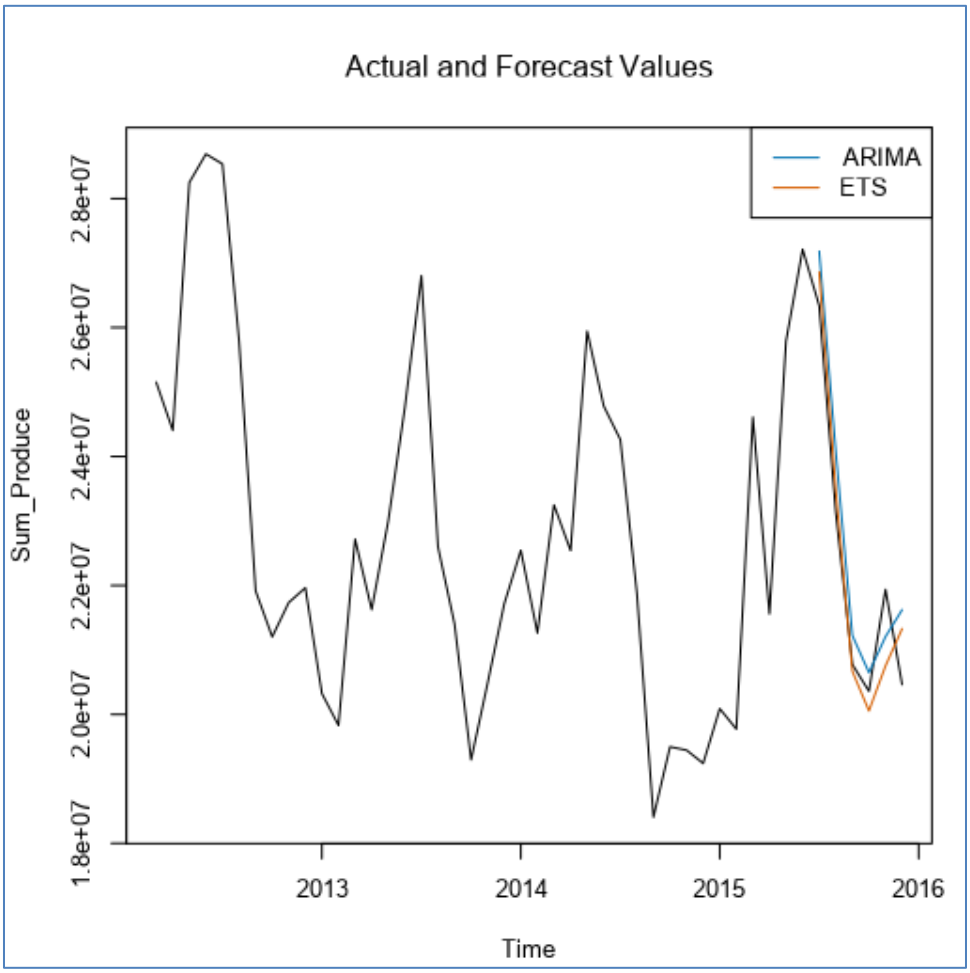
## Holdout Sample error measures

- When looking at the model's ability to predict the holdout sample, we see that the **ETS** model has **better predictive qualities** in just about **every metric**.

Accuracy Measures:

| Model | ME | RMSE | MAE | MPE | MAPE | MASE |
|---|---|---|---|---|---|---|
| ARIMA | -492238.83 | 792197.3 | 735878.2 | -2.1992 | 3.3098 | 0.433 |
| ETS | -21581.13 | 663707.2 | 553511.5 | -0.0437 | 2.5135 | 0.3257 |

- So, we will use the **ETS Model** for Forecasting.

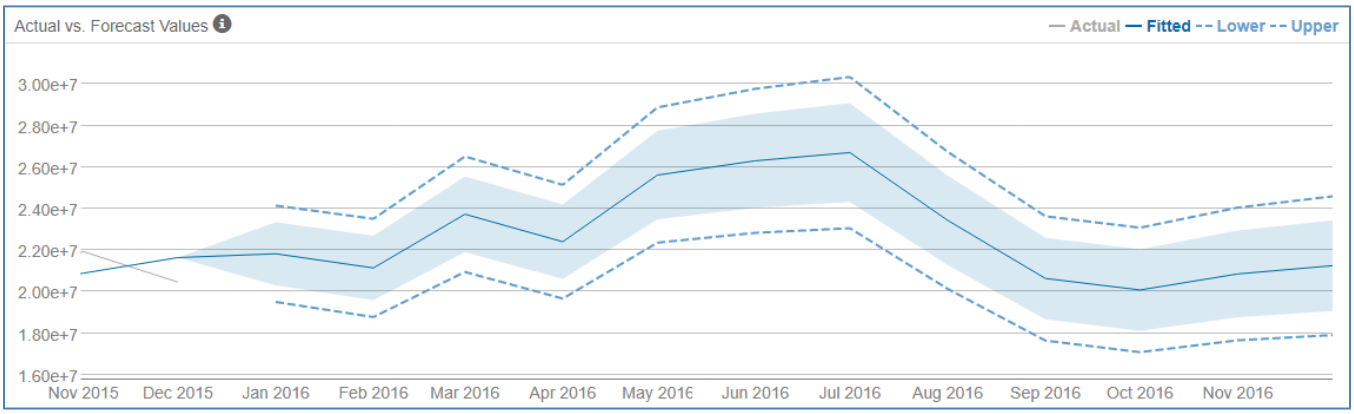


## *Existing Stores*

- Forecast results using 95% and 80% confidence intervals:

| Period | Sub_Period | Produce_Monthly_Sales | Produce_Monthly_Sales_high_95 | Produce_Monthly_Sales_high_80 | Produce_Monthly_Sales_low_80 | Produce_Monthly_Sales_low_95 |
|---|---|---|---|---|---|---|
| 2016 | 1 | 21829060.031666 | 24149899.115321 | 23346575.14138 | 20311544.921952 | 19508220.948011 |
| 2016 | 2 | 21146329.631982 | 23512577.365832 | 22693535.862148 | 19599123.401815 | 18780081.898131 |
| 2016 | 3 | 23735686.93879 | 26517865.796798 | 25554855.912929 | 21916517.964651 | 20953508.080782 |
| 2016 | 4 | 22409515.284474 | 25150243.401256 | 24201581.075733 | 20617449.493214 | 19668787.167691 |
| 2016 | 5 | 25621828.725097 | 28880596.484529 | 27752622.431914 | 23491035.018279 | 22363060.965665 |
| 2016 | 6 | 26307858.040046 | 29777680.067343 | 28576652.715009 | 24039063.365084 | 22838036.01275 |
| 2016 | 7 | 26705092.556349 | 30348682.320364 | 29087507.847195 | 24322677.265503 | 23061502.792334 |
| 2016 | 8 | 23440761.329527 | 26742106.733295 | 25599395.061562 | 21282127.597491 | 20139415.925758 |
| 2016 | 9 | 20640047.319971 | 23635033.372194 | 22598363.439189 | 18681731.200753 | 17645061.267747 |
| 2016 | 10 | 20086270.462075 | 23084199.797487 | 22046511.090727 | 18126029.833423 | 17088341.126662 |
| 2016 | 11 | 20858119.95754 | 24055437.105831 | 22948733.269445 | 18767506.645635 | 17660802.809249 |
| 2016 | 12 | 21255190.244976 | 24596988.126893 | 23440274.43075 | 19070106.059202 | 17913392.363058 |

- Shown graphically here:



## *New Stores*

We will be forecasting for **each of the clusters** and then **multiplying the results by the number of new stores in that cluster**.

Then we will be adding all of these forecasts together **on the same months** to get a **total forecast for all the new stores**.

| Record | Period | Sub_Period | New_Stores |
|---|---|---|---|
| 1 | 2,016 | 1 | 2,563,357.910041 |
| 2 | 2,016 | 2 | 2,483,924.727562 |
| 3 | 2,016 | 3 | 2,910,944.145687 |
| 4 | 2,016 | 4 | 2,764,881.869697 |
| 5 | 2,016 | 5 | 3,141,305.867305 |
| 6 | 2,016 | 6 | 3,195,054.203804 |
| 7 | 2,016 | 7 | 3,212,390.95409 |
| 8 | 2,016 | 8 | 2,852,385.769198 |
| 9 | 2,016 | 9 | 2,521,697.18679 |
| 10 | 2,016 | 10 | 2,466,750.893696 |
| 11 | 2,016 | 11 | 2,557,744.587714 |
| 12 | 2,016 | 12 | 2,530,510.805133 |

**Before you submit**

Please check your answers against the requirements of the project dictated by the rubric. Reviewers will use this rubric to grade your project.

*Thank You*